From Customer Lifetime Value Ranking to Uplift-Aware Targeting: Cross-Context Evidence from Online Retail, Grocery, and Supermarket Campaign Data

Vu Minh Anh Nguyen[1] and Linh Hoang Nguyen[1*]

[1*]Department of Information Technology, FPT University, Ho Chi Minh City, Vietnam.

*Corresponding author(s). E-mail(s): linhnh67@fpt.edu.vn; Contributing authors: anhnvmse203412@fpt.edu.vn;

Abstract

Customer lifetime value (CLV) models identify *who is valuable*; uplift models identify *who is responsive* to marketing. These objectives are seldom studied together or across heterogeneous retail contexts. This work connects explainable CLV ranking with uplift-aware targeting on three datasets spanning UK e-commerce, US grocery, and a randomised supermarket SMS campaign. First, under a walk-forward protocol aligned with prior work, a Hurdle–Semantic CLV framework remains competitive across the two CLV contexts, though the dominant mechanism shifts – e-commerce is incidence-dominated, grocery is sequence-dominated – and a leave-one-group-out ablation shows sequence features significantly help grocery while semantic profiles mainly aid interpretation. Second, on the campaign data, ranking by predicted *response* is *worse than random* for incremental targeting, whereas uplift learners recover persuadable customers. Third, value and persuadability select almost disjoint customers (top-decile overlap **0.7%**), and a *value-adjusted uplift* score ($\hat{\tau}_i \cdot \hat{V}_i$) achieves the highest point estimate of profit under tight budgets, although these monetary differences do not reach statistical significance. In short: CLV identifies who is valuable, uplift identifies who is responsive, and value-adjusted uplift identifies who is worth targeting.

Keywords: customer lifetime value, uplift modelling, explainability, conformal prediction, cross-context generalisation, campaign targeting

1 Introduction

Customer lifetime value (CLV) prediction and uplift modelling are two mature but largely separate strands of marketing analytics. CLV studies rank customers by expected future value and direct retention budgets towards the high-value tail. Uplift studies, by contrast, estimate the *incremental* effect of a marketing action and direct campaign budgets towards customers whose behaviour actually changes when treated. The two objectives can disagree sharply: a loyal high-value shopper who would return anyway yields little incremental revenue if targeted, whereas a mid-value customer who only re-purchases when prompted – the "persuadable" – is invisible to a value model but is exactly whom a campaign should reach.

Despite the practical importance of this distinction, three gaps remain. (i) CLV and uplift are rarely evaluated within a single, comparable pipeline. (ii) CLV frameworks are seldom stress-tested across structurally different retail contexts, so claims of generality are largely untested. (iii) Targeting studies frequently optimise either value or response in isolation, leaving the combined "who is both valuable and persuadable" question unanswered. These gaps are addressed with an explainable, uncertainty-aware framework evaluated across three datasets, and with an explicit *value-adjusted uplift* targeting rule.

The study is organised around four research questions:

- **RQ1** How robust is the explainable Hurdle–Semantic CLV framework when re-estimated across UK online retail and US grocery contexts?
- **RQ2** Do semantic product profiles and behavioural sequence features provide incremental value beyond RFM for CLV ranking, calibration, and interpretation?
- **RQ3** On campaign data with treatment/control labels, does uplift-aware targeting outperform CLV-only, RFM-only, and response-model targeting?
- **RQ4** Can combining predicted customer value with uplift scores improve simulated campaign efficiency, and what transferable design principles follow for budget-constrained retail campaigns?

Contributions. (1) A cross-context evaluation showing the Hurdle–Semantic framework stays competitive, with a leading mechanism that depends on context zero-inflation (RQ1). (2) A leave-one-group-out ablation isolating where each feature group helps – sequence in grocery, semantic mainly for interpretation – with SHAP and conformal analyses (RQ2). (3) Evidence that response-based targeting is *worse than random* for incremental campaigns (RQ3). (4) A value-adjusted uplift policy with a profit simulation and an honest account of when it helps (RQ4).

2 Related Work

2.1 CLV prediction and explainability

Customer-base analysis has a long lineage of probability models that estimate future value from RFM summaries [1, 2]. Retail CLV is typically zero-inflated and heavy-tailed, which motivates two-part Hurdle models and zero-inflated lognormal formulations that separate purchase incidence from spend magnitude [3, 4]. Beyond RFM, sequence and temporal features [5] and learned product embeddings [6] enrich tabular CLV models, commonly trained with gradient boosting [7]. For decision support, post-hoc attribution [8] and calibrated uncertainty are increasingly expected; conformalized quantile regression [9] yields intervals with valid marginal coverage under heteroscedasticity.

2.2 Uplift modelling and heterogeneous treatment effects

Uplift (true-lift) modelling estimates the incremental impact of a treatment rather than the response level [10]. Meta-learners (S-, T-, and X-Learners) and causal forests learn heterogeneous treatment effects with flexible base learners [11, 12]. Closest in motivation, the analysis of [13] shows that targeting the highest-risk (or highest-propensity) customers can be ineffective because they would act anyway – the empirical core of the "response $\neq$ persuadable" finding.

2.3 Profit-driven and value-based targeting

A separate strand argues that targeting should optimise profit, not response: profit-driven churn and campaign models [14–16], and policy evaluation that ranks targeting rules by incremental value [17]. This work bridges these strands by *combining* an explainable CLV value estimate with an uplift estimate into a single value-adjusted policy, and by evaluating that policy across structurally different retail contexts. Unlike that line of work – typically single-domain, black-box, and reporting point performance – the present study adds cross-context validation, SHAP-based interpretation, honest significance testing, and an explicit value-versus-persuadability overlap analysis. Methodologically, the contribution is not a new estimator but a value-adjusted *policy* ($\hat{\tau}_i \hat{V}_i$) evaluated head-to-head against CLV-only and uplift-only policies, with an overlap analysis that quantifies how differently they select customers.

3 Data and Problem Setup

Three datasets with deliberately separated roles are used. **Online Retail II** (UK e-commerce) is the baseline CLV context, reused read-only from prior work. **Dunnhumby Complete Journey** (US grocery; 2,500 households, 102 weeks, 2.6M lines) is an *external* CLV validation context. **X5 RetailHero** provides a randomised SMS campaign with treatment/control labels for 200,039 clients and 45.7M pre-communication purchase lines.

To keep the causal claim clean, coupon redemption in Dunnhumby – a *post-campaign outcome*, not a clean assignment – is *not* used as a treatment. Uplift estimation is restricted to the randomised X5 design (balanced, treatment ratio 0.500; raw uplift +3.32pp, 95% CI $[2.90, 3.75]$); Dunnhumby serves only CLV validation. The CLV target over a horizon is $Y_i = \sum_t \text{Revenue}_{i,t}$. The two CLV contexts differ sharply: Online Retail II has a $\sim 50\%$ zero rate and non-zero skewness ≈ 16.7 (sparse, heavy-tailed), whereas Dunnhumby has a $\sim 7\%$ zero rate and skewness ≈ 2.5 (dense, light-tailed) – a structural contrast that makes RQ1 a genuine robustness test.

4 Methodology

4.1 Module A — Hurdle CLV ranking

Value is decomposed into incidence and magnitude,

$$\widehat{CLV}_i = p_i \cdot m_i, \quad p_i = P(Y_i > 0 \mid X_i), \quad m_i = \mathbb{E}[Y_i \mid Y_i > 0, X_i], \tag{1}$$

where stage 1 is a gradient-boosted classifier and stage 2 a gradient-boosted regressor on $\log Y_i$ restricted to positive customers. A lognormal bias correction is applied, $m_i = \exp(\hat{\mu}_i + \hat{\sigma}^2/2)$, with $\hat{\sigma}^2$ the residual variance of stage 2. The decomposition is attractive when the target is a mixture of structural zeros and a heavy-tailed positive part.

4.2 Module B — Semantic product profiling

A basket-level product co-occurrence matrix is transformed by positive point-wise mutual information and factorised by truncated SVD into 32-dimensional product embeddings. Each customer's taste vector is the revenue-weighted mean of the embeddings of the products they bought, computed over the full observation window and the last 90 days to capture taste drift.

4.3 Module C — Uplift learners

With treatment $T_i \in \{0, 1\}$ and outcome Y_i, the uplift is $\tau_i = E[Y_i(1) - Y_i(0) \mid X_i]$. It is estimated with an S-Learner (a single model with treatment as a feature), a T-Learner (separate treated/control models), an X-Learner (a two-stage meta-learner that imputes individual effects and regresses on them), and a class-transformation model. These are compared with a response model (which ranks by $P(Y_i = 1)$ ignoring treatment) and a random baseline.

4.4 Module D — Value-adjusted uplift (central contribution)

The central targeting rule, and the main contribution of this work, scores customers by

$$\text{Score}_i^{\text{VU}} = \hat{\tau}_i \cdot \hat{V}_i, \tag{2}$$

where $\hat{V}_i$ is a value proxy derived from historical spend. Notably, $\hat{V}_i$ is a *proxy*, not an observed future CLV, and results are reported accordingly.

4.5 Evaluation protocol

CLV: walk-forward (temporal).

CLV is a forecasting task, so a random split would leak future structure. CLV is therefore evaluated with a *walk-forward* design. For Online Retail II, the Phase 1 walk-forward windows are reused (anchored 1 Dec 2009; benchmark windows W1–W3, with W4–W5 as extended robustness), using the same RFM and behavioural features and the same VIP label. This makes the Phase 2 numbers directly comparable to, and consistent with, Phase 1. For Dunnhumby, four week-level windows are built (observation→prediction: 26→8, 39→13, 52→13, 78→13 weeks). Within each window a stratified 80/20 split on the top-decile VIP indicator (seed 42) is used; Normalized Gini (NG), Revenue Capture@10% (RC@10), Precision@10%, Spearman, and decile calibration are reported as the *mean ± std across windows.*

Uplift: randomised cross-section.

The X5 campaign is a cross-sectional randomised design, so a 70/30 split stratified on the joint of treatment and outcome is used, reporting Qini AUC, AUUC, and uplift@K. Because treatment is randomised, any top-K set contains both treated and control customers, so incremental conversions and revenue are the difference in group means scaled by the number targeted. Uncertainty is quantified with the percentile bootstrap ($B = 1000$ for CLV, $B = 400$ for uplift), reporting paired differences with 95% confidence intervals.

4.6 Implementation details

For reproducibility the following choices are fixed. *Preprocessing.* Online Retail II drops cancelled invoices (invoice codes starting with `C`), non-positive quantity or price, and rows with a missing customer identifier; revenue is quantity×price, giving 805,549 transactions for 5,038 customers (matching Phase 1). Dunnhumby uses all 2,500 households with net sales value as revenue; coupon and promotion variables are *not* used as a treatment. X5 features are aggregated strictly from pre-communication purchases, and the outcome is the binary purchase-response label supplied by the RetailHero benchmark; no post-campaign revenue is released, hence the pre-period value proxy. *Models.* All gradient-boosted stages use LightGBM: the Hurdle classifier (400 trees, learning rate 0.03) and regressor (500 trees, 0.03); the uplift base learner is a LightGBM classifier (300 trees, 0.05, 31 leaves, subsample and colsample 0.8, seed 42). The S-, T-, and class-transformation learners use `scikit-uplift`; the class-transformation target is $z_i = Y_i T_i + (1-Y_i)(1-T_i)$ at propensity $\frac{1}{2}$, so $\hat{\tau}_i = 2\hat{P}(z_i{=}1) - 1$, and the X-Learner is implemented directly with LightGBM regressors averaged at propensity 0.5. Qini AUC, AUUC, and uplift@K use the `scikit-uplift` implementations. *Explainability and uncertainty.* SHAP uses TreeExplainer on the Stage-2 regressor evaluated on the test fold. CQR is split-conformal: an 80/20 train/test split, with the training fold further split 50/50 into proper-training and calibration; LightGBM quantile regressors at levels $\alpha/2$ and $1 - \alpha/2$ are conformalised by the calibration residual quantile $\hat{q}$, and empirical coverage is measured on the test fold at nominal 90% and 80%.

5 Results and Analysis

5.1 Cross-context CLV ranking (RQ1, RQ2)

Table 1 reports the walk-forward CLV benchmark for both contexts as mean ± std across temporal windows. On Online Retail II the Hurdle model attains the highest

Table 1 Walk-forward CLV benchmark. NG and RC@10 are mean ± std across temporal windows (Online Retail II on the Phase 1 windows W1–W3, Dunnhumby on D1–D4); Precision@10 is the across-window mean. Results are temporally valid and lie in the same range as the Phase 1 walk-forward benchmark (NG ≈ 0.83); small differences follow from the feature set and the window subset.

context	model	NG	RC@10	Prec@10
Online Retail II (W1-W3)	Hurdle	0.825 ± 0.050	59.037 ± 9.466	0.577
Online Retail II (W1-W3)	Monetary	0.822 ± 0.058	60.017 ± 8.404	0.586
Online Retail II (W1-W3)	RandomForest	0.805 ± 0.043	56.883 ± 7.203	0.558
Online Retail II (W1-W3)	Ridge	0.805 ± 0.038	56.193 ± 8.234	0.547
Online Retail II (W1-W3)	XGBoost	0.802 ± 0.044	56.807 ± 8.818	0.540
Online Retail II (W1-W3)	LightGBM	0.799 ± 0.052	57.170 ± 9.850	0.547
Dunnhumby (D1-D4)	RandomForest	0.862 ± 0.012	34.297 ± 0.623	0.723
Dunnhumby (D1-D4)	Ridge	0.858 ± 0.009	34.280 ± 0.452	0.703
Dunnhumby (D1-D4)	Hurdle	0.855 ± 0.014	34.280 ± 0.457	0.714
Dunnhumby (D1-D4)	XGBoost	0.854 ± 0.014	34.235 ± 0.923	0.738
Dunnhumby (D1-D4)	LightGBM	0.849 ± 0.009	34.270 ± 0.778	0.723
Dunnhumby (D1-D4)	Monetary	0.825 ± 0.010	32.350 ± 0.712	0.658

Table 2 X5 uplift model comparison (RQ3). The response model ranks below random.

model	Qini_AUC	AUUC	uplift@10	uplift@20
S-Learner	0.014	0.020	0.112	0.077
X-Learner	0.012	0.018	0.104	0.072
T-Learner	0.010	0.015	0.087	0.069
ClassTransform	0.008	0.011	0.086	0.064
Random	-0.002	-0.004	0.026	0.029
ResponseModel	-0.007	-0.011	0.004	0.008

point estimate (NG$= 0.825 \pm 0.050$, RC@10$=$ 59.0%) but is statistically indistinguishable from a strong Monetary baseline (NG$=$ 0.822, paired $p = 0.55$). On Dunnhumby it is again statistically tied with regularised linear and tree models (NG$\approx$ 0.855, within one standard deviation), and is not the top model. The framework thus remains *competitive* across contexts rather than dominant, while the leading mechanism shifts between them. These numbers are consistent with the Phase 1 result (Hurdle NG$\approx$ 0.83), replacing an earlier inflated random-split estimate. The mechanism is visible in the data structure: the $\sim$ 50% zero rate of e-commerce lets the Hurdle incidence stage carry ranking information, whereas the $\sim$ 7% zero rate of grocery makes the problem magnitude-dominated, where the two-part structure adds little.

For RQ2, a leave-one-group-out ablation is run under the walk-forward protocol (Table 5, with paired tests in Table 4). The honest picture is nuanced. Adding all feature groups beats RFM significantly in grocery (ΔNG$= +0.044$, $p = 0.002$, driven by sequence features) but *not* significantly in e-commerce (ΔNG$= +0.009$, $p = 0.17$). Dropping the semantic group does *not* significantly change ranking in either context (paired $p = 0.58$ and 0.84), whereas dropping sequence significantly hurts grocery ($p = 0.001$). The claim is therefore limited to what the data support: sequence features are a significant ranking driver in grocery, while semantic profiles are valuable

Table 3 Targeting policy comparison on X5 (RQ4). The value proxy $\hat{V}_i$ is pre-period historical monetary spend (a gross-value proxy, not future CLV). Incremental conversions and incremental value are held-out treated-minus-control estimates within each selected top-K set (Eq. 3); the headline uses margin $m = 1$, so profit = (incr. value) − contact cost with $c = 100$ rubles per contact (value and profit in millions of rubles). Across a sensitivity sweep ($m \in \{0.1, 0.2, 0.3\}$, $c \in \{50, 100, 200\}$) value-adjusted uplift has the highest point-estimate profit in all nine settings.

policy	K	incr. conv.	incr. value (M)	profit (M)
Random	10%	129.4	0.29	-0.31
Random	20%	351.5	0.13	-1.07
RFM-only	10%	52.0	0.11	-0.49
RFM-only	20%	109.7	1.96	0.76
Value-only(CLV proxy)	10%	77.3	-0.05	-0.65
Value-only(CLV proxy)	20%	263.0	4.56	3.36
Uplift-only(S-Learner)	10%	670.4	1.09	0.49
Uplift-only(S-Learner)	20%	930.1	3.23	2.03
Value-adjusted(uplift x value)	10%	248.4	1.34	0.74
Value-adjusted(uplift x value)	20%	451.7	5.23	4.03

Table 4 Significance of key claims (ΔNG with walk-forward paired t-tests for CLV; bootstrap for the uplift Qini comparison). With only 3–4 windows, the CLV tests are robustness diagnostics rather than definitive inference. Only sequence-in-grocery and uplift-vs-response are significant.

claim	ΔNG	p
Hurdle > Monetary (e-com, walk-fwd)	+0.0030	0.554
Hurdle vs RandomForest (grocery, walk-fwd)	−0.0064	0.063
extra features > RFM (e-com)	+0.0092	0.1668
extra features > RFM (grocery)	+0.0444	0.0021
semantic adds ranking (e-com)	+0.0028	0.5835
semantic adds ranking (grocery)	+0.0011	0.8373
sequence adds ranking (grocery)	+0.0206	0.0011
S-Learner > Response (uplift Qini)	–	< 0.001

for interpretation and segmentation rather than as a stand-alone ranking booster – consistent with the Phase 1 finding. The ALL row in Table 5 differs slightly from the Hurdle row in Table 1 because the former uses the full feature-group set whereas the latter reports the base benchmark configuration. Stage-2 SHAP attribution supports this reading (Fig. 1): monetary value dominates conditional spending in e-commerce, whereas recent-sequence spend dominates in grocery. Conformalized quantile regression (Table 7) attains valid coverage in both contexts (90% → 90.2%/89.0%), with substantially tighter intervals in grocery, consistent with its lighter tail.

5.2 Uplift-aware targeting (RQ3)

The campaign is a balanced design: across all pre-communication covariates the standardised mean difference between treated and control is below 0.01, so group-mean differences provide a low-bias estimate of incremental effects. Two evaluation regimes

Table 5 Leave-one-group-out ablation (Hurdle, walk-forward). dNG_vs_ALL > 0 means dropping the group hurts; paired_p is the paired t-test across windows.

context	variant	NG_mean	NG_std	dNG_vs_ALL	paired_p
Online Retail II	ALL	0.829	0.056	0.000	–
Online Retail II	ALL-behavioural	0.829	0.052	-0.000	0.954
Online Retail II	ALL-semantic	0.826	0.048	0.003	0.584
Online Retail II	RFM-only	0.820	0.063	0.009	0.167
Dunnhumby	ALL	0.856	0.015	0.000	–
Dunnhumby	ALL-behavioural	0.856	0.015	0.000	0.088
Dunnhumby	ALL-semantic	0.855	0.016	0.001	0.837
Dunnhumby	ALL-sequence	0.836	0.017	0.021	0.001
Dunnhumby	RFM-only	0.812	0.013	0.044	0.002

Table 6 Top-10% targeting overlap between policies, defined as $|A_K \cap B_K|/K$ where A_K, B_K are the top-K customer sets selected by each policy. Value and Uplift select almost disjoint customers (0.007), evidencing that value $\neq$ persuadability.

policy_A	policy_B	overlap
RFM	Value(CLV proxy)	0.666
RFM	Uplift(S-Learner)	0.003
RFM	Value-adjusted	0.173
Value(CLV proxy)	Uplift(S-Learner)	0.007
Value(CLV proxy)	Value-adjusted	0.312
Uplift(S-Learner)	Value-adjusted	0.256

Table 7 Conformalized quantile regression: empirical coverage and mean interval width per context. Coverage is close to nominal; grocery intervals are tighter, consistent with its lighter tail.

context	nominal_coverage	empirical_coverage	mean_interval_width
Online Retail II	0.900	0.902	1,653
Online Retail II	0.800	0.793	1,315
Dunnhumby	0.900	0.890	707
Dunnhumby	0.800	0.778	515

are kept distinct: ranking quality (NG, RC@10, Qini) measures how well a score orders customers, whereas the profit simulation evaluates the held-out economic outcome of acting on that score under a budget. Table 2 delivers the central uplift result: the response model ranks *below random* (Qini -0.007; its bootstrap CI is entirely negative, Table 4), while uplift learners achieve positive Qini and reach $\approx 11\%$ uplift in the top decile – more than three times the average treatment effect. Ranking by "who is likely to buy" actively selects sure buyers and therefore wastes budget; ranking by estimated incremental effect recovers the persuadable segment. The paired bootstrap difference between the best uplift learner and the response model is significant ($p < 0.001$).

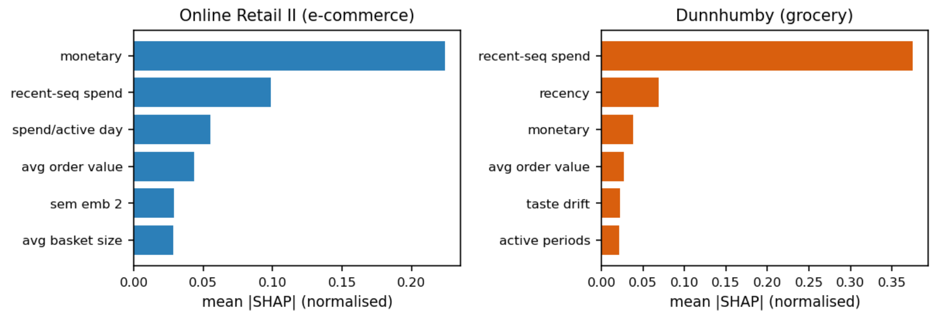

Fig. 1 Stage-2 SHAP attribution (Hurdle model). In e-commerce, monetary value dominates conditional spending; in grocery, recent-sequence spend is the leading driver. This supports the context-dependent mechanism shift (RQ2).

5.3 Value-adjusted targeting and profit (RQ4)

A first finding is that value and persuadability identify different customers: the top-10% sets chosen by a value (CLV-proxy) policy and an uplift policy overlap by only 0.007 (Table 6). Value-adjusted uplift, by construction, blends the two (overlap 0.31 with value and 0.26 with uplift). Each policy scores every customer and selects the top-K set A_K; profit is then evaluated on the held-out RCT as

$$\widehat{\text{Profit}}@K = m\,\widehat{\Delta V}_K - c\,|A_K|, \qquad \widehat{\Delta V}_K = |A_K| \left(\overline{Y_i \hat{V}_i}\big|_{T=1,\,A_K} - \overline{Y_i \hat{V}_i}\big|_{T=0,\,A_K} \right), \quad (3)$$

where Y_i is the observed campaign outcome, $\hat{V}_i$ the pre-communication historical-monetary value proxy (in rubles), m a gross margin, and c the per-contact cost. Crucially, the policy score $\hat{\tau}_i \hat{V}_i$ is used *only* to select A_K; profit is measured from the treated-minus-control outcome difference *within* A_K, so it is a held-out estimate and is not mechanically implied by the score. Because X5 releases a binary purchase outcome but no post-campaign revenue, $\widehat{\Delta V}_K$ is a value-weighted incremental-conversion estimate (an expected value under the proxy $\hat{V}_i$) rather than observed revenue; profit is therefore an expected, not observed, quantity and is reported with the hedge below. Table 3 compares five policies by incremental conversions, incremental gross value, and profit (margin $m = 1$ in the headline). Targeting by value alone can *lose* money at small budgets, because the highest-value customers are disproportionately sure buyers with low uplift; uplift-only targeting maximises the *number* of incremental conversions; and value-adjusted uplift achieves the highest point estimate of profit at tight budgets (Fig. 2) by concentrating spend on high-value persuadables. This ranking is stable: in a sensitivity sweep over $c \in \{50, 100, 200\}$ and $m \in \{0.1, 0.2, 0.3\}$, value-adjusted uplift remains the top policy by this point estimate across all nine settings. Crucially, however, the bootstrap analysis (Table 4) shows that while the *ranking*-level claims (uplift vs response; sequence in grocery) are significant, the monetary profit differences between policies are directional rather than statistically significant given test-set noise. This is stated explicitly rather than overclaimed.

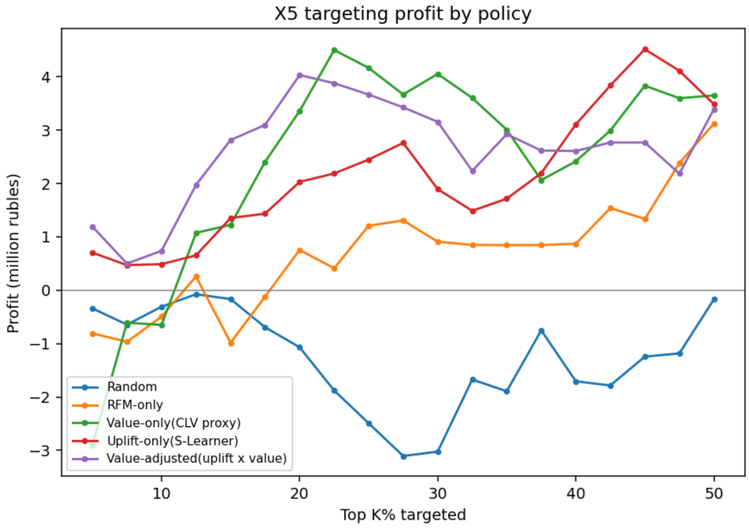

Fig. 2 Simulated targeting profit by policy and budget size (RQ4). Value-adjusted uplift achieves the highest point-estimate profit at tight budgets; value-only targeting can be unprofitable.

6 Discussion

6.1 Why CLV and uplift disagree

The disagreement is not marginal but near-total: the value and uplift policies share only 0.7% of their top decile (Table 6). The mechanism is selection on different quantities – value ranks on expected spend, which is highest for already-loyal customers, whereas uplift ranks on the *derivative* of spend with respect to treatment, which is highest for marginal customers. This is the empirical counterpart of the "retention futility" argument [13]: the most valuable or most likely customers are often the least movable, which is why value-only targeting can be unprofitable at tight budgets. Value-adjusted uplift resolves the tension by multiplying the two signals, recovering high-value *persuadables*.

6.2 Why the CLV mechanism shifts across contexts

The two retail regimes differ in their zero-inflation ($\sim$ 50% vs $\sim$ 7% zero rate): e-commerce is incidence-dominated, so the Hurdle stage-1 carries information and the model is competitive; grocery is magnitude-dominated and habitual, so recent-sequence behaviour is the significant driver and a two-part decomposition adds little. The framework therefore *transfers*, but which component does the work is context-specific – a caution against assuming a single architecture is universally best.

6.3 Design principles

Several principles follow, offered as transferable guidance rather than numbers: targeting on value alone selects sure buyers and can lose money at tight budgets; targeting on response alone can underperform random; value-adjusted uplift suits budget-constrained settings; explainability clarifies *why* a customer is selected; and calibrated uncertainty matters when data are unstable. No direct numerical transfer to a specific market is claimed.

6.4 Limitations

Several limitations bound these claims. The Dunnhumby CLV sample is modest ($\sim$ 2,500 households), and the uplift evidence comes from a single SMS campaign in one market and period, so external validity is untested. The X5 value $\hat{V}_i$ is pre-period historical spend, a proxy for future value rather than an observed future CLV. Monetary profit differences between policies are directional, not statistically significant, and absolute Qini values are modest because the raw campaign signal is small (≈ 3.3 percentage points); the walk-forward CLV tests use only 3–4 windows, limiting paired-test power. Uplift is identified only on the randomised X5 design; no causal claim is made on the observational CLV datasets.

7 Conclusion and Future Work

Across three retail contexts, this work connects explainable CLV ranking with uplift-aware targeting. The Hurdle–Semantic framework remains competitive across contexts, but the dominant mechanism is context-dependent. The additional feature groups improve ranking mainly in grocery, driven by sequence features, while semantic profiles aid interpretation more than aggregate ranking. On the campaign data, response-based targeting is worse than random for incremental campaigns, and value-adjusted uplift ranks first on point-estimate profit under tight budgets, though these monetary differences are not statistically significant. CLV identifies who is valuable; uplift identifies who is responsive; value-adjusted uplift identifies who is worth targeting. Future work includes jointly learned value-and-uplift objectives, multi-campaign and multi-market replication, and a field test of the value-adjusted policy under a real budget constraint.

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